

• Set identities can also be proved using membership tables. We consider each combination of the atomic sets (that is, the original sets used to produce the sets on each side) that an element can belong to and verify that elements in the same combinations of sets belong to both the sets in the identity.

To indicate that an element is in a set, a 1 used; to indicate that an element is not in a set, a 0 is used.

Example 13: Use a membership table to show that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Solution:

A	B	C	$B \cup C$	$A \cap (B \cup C)$	$A \cap B$	$A \cap C$	$(A \cap B) \cup (A \cap C)$
1	1	1	1	1	1	1	1
1	1	0	1	1	1	0	1
1	0	1	1	1	0	1	1
1	0	0	0	0	0	0	0
0	1	1	1	0	1	1	1
0	1	0	1	0	1	0	1
0	0	1	1	0	0	1	1
0	0	0	0	0	0	0	0

• Once we have proved set identities, we can use them to prove new identities. In particular, we can apply a string of identities, one in each step, to take us from one side of a desired identity to the other.

Example 14: Let A, B and C be sets. Show that $\overline{A \cap (B \cap C)} = (\overline{A} \cup \overline{B}) \cap \overline{A}$.

Solution:

$$\begin{aligned}
 \overline{A \cap (B \cap C)} &= \overline{A \cap \overline{B \cap C}} \quad (\text{2nd De Morgan's Law}) \\
 &= \overline{A \cap (\overline{B} \cup \overline{C})} \quad (\text{1st De Morgan's Law}) \\
 &= (\overline{A \cap \overline{B}}) \cap \overline{A} \quad (\text{2nd Commutative Law}) \\
 &= (\overline{A} \cup \overline{B}) \cap \overline{A} \quad (\text{1st Commutative Law}) \quad \square
 \end{aligned}$$

2.2.3 Generalized Unions and Intersections

Because unions and intersections of sets satisfy associative laws, the sets $A \cup B \cup C$ and $A \cap B \cap C$ are well defined; that is the meaning of this notation is unambiguous when A, B and C are sets.

• Note that $A \cup B \cup C$ contains those elements that are in at least one of the sets A, B and C, and that $A \cap B \cap C$ contains those elements that are in all of A, B and C. $\square$